Riya Agrawal

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Education

Northwestern University, Evanston, IL — September 2025 – December 2026 (expected)
M.S. in Mechanical Engineering (Thesis), Biosystems

University of Illinois Urbana-Champaign, Champaign, IL — August 2021 – May 2025
B.S. in Mechanical Engineering

Skills

- **Design:** SolidWorks, Fusion 360, Onshape, Creo
- **Simulation & Analysis:** Abaqus, Ansys, OpenSim, CoppeliaSim
- **Fabrication & Testing:** Instron mechanical testing, CNC mill and lathe, 3D printing, carbon fiber layup
- **Hardware & Programming:** Arduino, Raspberry Pi Pico, STM32, PCB design, C, C++, Python, MATLAB

Research Experience

Research Assistant — July 2025 - Present
Northwestern University Prosthetics-Orthotics Center – Chicago, IL

- Led design and prototyping of adjustable-stiffness passive dynamic ankle-foot orthotic struts that exceed the range of motion of commercial struts
- Verified designs through FEA in Abaqus, evaluating deformation under physiological loading to validate structural performance before manufacturing
- Machined aluminum and steel prototypes on the mill and lathe and ran Instron testing under varied loading, correlating rotational stiffness with FEA predictions and published literature targets
- Conducted user-performance testing through human subject gait analysis in a simulated walking environment, comparing outcomes with normative gait data

Projects

Haptic Paddle & Embedded Mechatronics — March - June 2026

- Built a haptic paddle with programmable force feedback using a DRV8833 H-bridge, load cell, and AS5600 encoder
- Programmed Raspberry Pi Pico and STM32 microcontrollers in C/CircuitPython, implementing I^2C/SPI communication, PWM motor actuation, and current sensing for real-time control
- Designed force–displacement profiles simulating bumps, detents, and toggles with Python visualization

Expanding Heart Conduit (Senior Design) — January - May 2025

- Designed a balloon-expandable RV–PA conduit for pediatric heart patients to reduce repeat surgeries
- Modeled and optimized expanding stent geometries under cardiac loading using FEA in Abaqus
- Assessed candidate biomaterials for fatigue resistance, elastic recovery, manufacturability, and biocompatibility

MoneyTip, Google Hardware Product Sprint — May - August 2022

- Conceptualized and engineered a device that delivers money-saving tips at the click of a button
- Designed an injection-moldable enclosure in Fusion 360 around Google's AIY Voice Kit and a Raspberry Pi

GoBabyGo Volunteer, Shirley Ryan AbilityLab — July 2026

- Engineered low-cost adaptive ride-on vehicles with seating and support systems tailored to each child's mobility needs

Leadership & Work Experience

External Vice President — August 2024 - May 2025
Women in Mechanical Science and Engineering (WiM), UIUC

- Managed corporate relations and organized professional development events (previously Publicity Director)

ECS Peer, Engineering Career Services — August 2024 - May 2025
Grainger College of Engineering, UIUC – Champaign, IL

- Critiqued over 200 resumes with targeted feedback and conducted mock technical and behavioral interviews